AP025252.1:906243-927440
Bryobacterales bacterium F-183

? Lipase_C Lipase_2 AP_endonuc_2 SIS SIS Fic_N Fic HTH_24 OMP_b-brl_4 CarboxypepD_reg RecA_C RecA RelE MqsA_antitoxin HSP70 HSP70 OMP_b-brl_4 Plug CarboxypepD_reg MFS_1 ? ? OxoGdeHyase_C Transket_pyr E1_dh 2-oxoacid_dh 2-oxogl_dehyd_N

CACVCF010000035.1:92140-100218

Candidatus Terasakiella magnetica

Lyase_1 ASL_C2 DUF2786 DUF7168 DUF3572 ? ? ? MqsA_antitoxin RelE ? ? DUF6626 ? ?

BK015216.1:9050-19650
Siphoviridae sp. ctpbe1

RelE HTH_3 Peptidase_S49 GpA_ATPase GpA_nuclease gpW Phage_portal_2 HDPD Phage_cap_E ParB_N ParB_N Methyltransf_23 ? ? ? ? Phage_attach

AAEDWH010000057.1:387-3594
Salmonella enterica

RelE ? RICTOR_M ? ? ?

AAFNMU010000040.1:26950-10289
Salmonella enterica subsp. enterica serovar Ouakam

N6_Mtase ? DUF6015 ? ANKs ANKs ? MqsA_antitoxin RelE ? ArdcN MPTase-PolyVal ? ? ? LPD7 CbiA

KI530719.1:1-2661
Acinetobacter nectaris CIP 110549

HAD HTH_3 RelE Cep57_CLD Rep3_N Rep3_C

DF820463.1:1008823-1026149

Candidatus Vecturithrix granuli

SBP_bac_1 DUF5110 Glyco_hydro_31_3rd Glyco_hydro_31_2nd Gal_mutarotas_2 ParE_like ? ANKs ANKs PIN HTH_3 RelE ? HlyD_3 ABC_tran FtsX B12-binding_2 B12-binding DUF1156 N6_N4_Mtase N6_N4_Mtase ?

BEXT01000001.1:3283958-3303083
Desulfonema ishimotonii

Lip_prot_lig_C LplA-B_cat Sacchrp_dh_NADP Sacchrp_dh_C DAHP_snth_FXD DAHP_synth_1 ? ? ? ? RelE HTH_3 ? Ligase_CoA_2 Succ_CoA_lig CoA_binding_2 ATP-grasp_5 ? PTS_EIIA_2 Glycos_transf_2 NMT1_3

BNTY01000002.1:4428-18361
Betaproteobacteria bacterium

ABC_tran FecCD ? Peripla_BP_2 Plug TonB_dep_Rec_b-barrel HTH_Crp_2 cNMPBD HTH_31 RelE ? DUF4760 FTR1 TgpA_N Transglutaminase iHEPN-DUF4129 T2SSG Cytochrome_C554 ?

BPKG01000001.1:297106-311661
Burkholderiales bacterium

? DUF3987 Toprim_3 Phage_AlpA wHTH-Calcineurin_assc Phage_int_M Phage_integrase HTH_3 RelE TPR_Slam SlipAM TbpB_B_D PTPS Radical_SAM HTH_92 ACT_9 ACT AA_kinase DUF6647

BQJW01000520.1:79-1754
Ardenticatenaceae bacterium

HTH_3 RelE Sulfotransfer_1

DMQI01000127.1:799-6428
Candidatus Competibacteraceae bacterium

TetR_C_13 TetR_N HTH_Tnp_1 HTH_3 RelE YdjM AhpC-TSA Thioredoxin_7

DNMA01000116.1:1-9260

Legionellales bacterium

AMP-binding ACAS_N RelE HTH_3 GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C BipA_C NAD_kinase NAD_kinase_C SMC_N RsmJ TatC UPF0738

DPRR01000029.1:155907-170832
Ignavibacteriales bacterium

Complex1_30kDa Complex1_49kDa 2Fe-2S_thioredx NADH_4Fe-4S SLBB Complex1_51K ? SpoIIIAC HTH_3 RelE Ferritin Gp_dh_C Gp_dh_N PGK DUF6576 Rhomboid DUF4296 DEAD Helicase_C RecQ_Zn_bind HRDC HTH_40 Por_Secre_tail Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10

DASSSG010000179.1:1-3550
Bryobacteraceae bacterium

? HTH_3 RelE ? Spermine_synth

DASYHS010000164.1:9712-18368
Arenibaculum sp.

DUF1272 Arm-DNA-bind_3 Phage_integrase AbiEii AbiEi_4 ? HTH_3 RelE RHH_4 ? HTH_31 HTH_26 KfrA_N ? HTH_36

DASYUU010000318.1:6522-17706
Candidatus Binatia bacterium

? ProQ dnstrm_HI1420 Gp49 TumE HVO_A0114 HTH_3 RelE NERD ClpB_D2-small AAA_2 TraG_N DUF1932 NAD_binding_2 Abhydrolase_6 Bac_luciferase ?

DRCH01000162.1:241-3998
Acidiferrobacteraceae bacterium

HTH_3 RelE HlyIII ?

DATAIA010000604.1:5085-12292
Xanthobacteraceae bacterium

YkuD DUF2076 YdjC Glycos_transf_2 GtrA_DPMS_TM HisG tRNA-synt_His tRNA-synt_His RelE

DATAVZ010000021.1:49615-59834
Terracidiphilus sp.

? Pro_CA SRP54_N SRP54 SRP_SPB AAA_15 AAA_15 ? AhpC-TSA HSP70 RelE HTH_3 DnaJ DnaJ_C BrnT_toxin

DATELY010000192.1:44-4492
Steroidobacteraceae bacterium

HTH_3 RelE SGSH_C Sulfatase Sulfatase

DATEGF010000276.1:6010-17544
Stellaceae bacterium

AICARFT_IMPCHas MGS adh_short_C2 ? GerE Receiver ? Peptidase_S15 MqsA_antitoxin RelE HTH_3 ? N6_N4_Mtase PhdYeFM_antitox PIN Terminase_2 PhdYeFM_antitox

DATEBS010000023.1:8180-19674
Gammaproteobacteria bacterium

YoeB_toxin Cys_Met_Meta_PP ANKs ? ? YhcG_C MqsA_antitoxin RelE Phage_int_SAM_1 Phage_integrase Phage_integrase Phage_integrase Phage_int_SAM_1 DUF1016_N YhcG_C ? FANCI_S3

DATFYQ010000194.1:1-4289
Terriglobia bacterium

? Glyco_hydro_42M Glyco_hydro_42 ? RelE HTH_3

DATFYR010000354.1:527-7034
Terracidiphilus sp.

Phage_integrase CMD HAD_2 PMT_2 CAP Metallophos_2 RelE HTH_31 TIR

DAOQNI010000733.1:1-2796
Anaerolineales bacterium

Penicil_amidase RelE HTH_3 Methyltransf_25

DAONGM010000026.1:11485-7538
Pontiellaceae bacterium

? RelE HTH_3 SOUL tRNA_Me_trans_C tRNA_Me_trans_M tRNA_Me_trans BCCT IGPS RF-1 PCRF CN_hydrolase LNT_N

DAORSV010000091.1:1-3660
Gammaproteobacteria bacterium

? RelE HTH_31 ? Phage_integrase KfrA_N CENP-F_leu_zip

DAOMUI010000158.1:2420-5827
Acinetobacter johnsonii

HTH_3 IstB_IS21 HTH_7 rve ? MqsA_antitoxin RelE

DAOTFS010000027.1:14388-11445
Hyphomicrobiaceae bacterium

DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase ? RelE MqsA_antitoxin ? Pantoate_ligase PRODH Pro_dh-DNA_bdg Pro_dh Aldedh ? DnaJ SPOR Peptidase_S11

DATJFK010000054.1:1144-3559
Gammaproteobacteria bacterium

? MqsA_antitoxin RelE GatB_N GatB_Yqey

DATLUI010000146.1:234-8368
Gammaproteobacteria bacterium

RelE HTH_3 AEP-TOTE ResIII Helicase_C ? GAF Queuosine_synth TGT YajC

DASZVQ010000192.1:24155-36390
Rhizomicrobium sp.

PP-binding adh_short_C2 Acyl_transf_1 MFS_1 MFS_1 ? MqsA_antitoxin RelE Ribosomal_S6 Ribosomal_L9_C Ribosomal_L9_N ? Glyoxalase DLH YwiC Sulfotransfer_2

DASZQP010000073.1:9937-24160
Rhizomicrobium sp.

? ? ? Aminotran_3 AlaDh_PNT_N AlaDh_PNT_C adh_short_C2 OmpA RelE HTH_31 DAHP_synth_2 ? zf-CHY ZnR Peptidase_S10 ADH_zinc_N ADH_N_2 GFO_IDH_MocA GFO_IDH_MocA_C3

DASZQC010000610.1:14-5750
Dongiaceae bacterium

Pyrophosphatase APH DUF6285 RelE MqsA_antitoxin SBP_bac_3 BPD_transp_1 SNase

DATMGL010000007.1:45690-9878
Sphingobium sp.

DUF3363 RelE HTH_3 Transposase_20 DEDD_Tnp_IS110 AAA_15 ABC-ATPase UvrD-helicase UvrD_C_2 ? ParE_toxin Phage_integrase

DASPMZ010000809.1:1-6627
Reyranella sp.

BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran RelE HTH_3 BPD_transp_1 BPD_transp_1 BPD_transp_1_N oligo_HPY ABC_tran oligo_HPY ABC_tran

DASPLF010000069.1:7952-14693
Alphaproteobacteria bacterium

ThiS ? Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 CDP-OH_P_transf GIY-YIG UVR UvrC_RNaseH UvrC_RNaseH_dom HHH_5 NUDIX HTH_31 RelE

DASQLL010000409.1:1-15826
Pyrinomonadaceae bacterium

HSP70 DnaJ DnaJ_C ? TPRs TPRs TPRs FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 UnbV_ASPIC RelE ? CN_hydrolase CarboxypepD_reg OMP_b-brl_4

RHKO01000023.1:22971-35654
candidate division KSB1 bacterium

MFS_1 LPG_synthase_TM HMA ? ? Alginate_exp RelE HTH_3 His-Kinase His-Kinase HAMP DUF4390 Omp85 POTRA DUF4351 ? HicB_lk_antitox UPF0047

QQUM01000798.1:1-3471
Calditrichaeota bacterium

PhdYeFM_antitox YoeB_toxin RelE HTH_3 Ppx-GppA RVT_1

LDXN01000003.1:76731-82764
Candidatus Dadabacteria bacterium CSP1-2

ATP-synt_C ? tRNA_NucTran2_2 HD PolyA_pol_RNAbd PolyA_pol AMP-binding ? ParE_toxin ? RelE HTH_3 ?

NZUW01000061.1:149211-160336
Alcanivorax sp.

UCR_Fe-S_N Rieske Ribosomal_S9 Ribosomal_L13 Aldo_ket_red ? Lipase_GDSL_2 HTH_3 RelE SymE_toxin DUF6777 Toluene_X Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 AFG1_ATPase Abhydrolase_1 DLH ZapG-like

JACCXJ010000253.1:27561-40176
Gammaproteobacteria bacterium

SLATT_2 CinA DDE_Tnp_ISL3 ? Thioredox_DsbH Glyco_hydro_88 HNH_5 GmrSD_N RelE HTH_3 GFA Glyoxalase_6 Y2_Tnp HigB-like_toxin HTH_3 ?

JACDDO010000119.1:735-11630
Acidobacteria bacterium

TonB_C Sec_GG SecD_SecF_C Sec_GG SecD_1st SecDF_P1_head SecD_SecF_C YajC TGT MauE AhpC-TSA AhpC-TSA ? RelE MMR_HSR1 MMR_HSR1 KH_dom-like P_proprotein P_proprotein

JABXPR010000001.1:751232-767756
Citrobacter sp. RHBSTW-00671

? RsmJ Peptidase_M3 TOP_N SAM_MT Pyr_redox_2 ? HTH_3 RelE ? Usp UspB PHO4 HI0933_like FCD GntR Mannitol_dh Mannitol_dh_C

JACMKT010000016.1:328-14089
Rhodospirillales bacterium

Cupin_2 DUF4384 DUF4399 DUF4399 Caspase Big Big FGS HTH_Crp_2 cNMPBD HTH_31 RelE CHASE2 Guanylate_cyc OmpA adh_short_C2 Peptidase_S49 Peptidase_S49 ? ? Chorismate_synt

JACMKN010000103.1:78-8262
Deltaproteobacteria bacterium

DUF4160 HTH_3 DUF2442 HTH_3 RelE PEP-CTERM fn3 Big Big_3_3 Big Receiver NtrC-AAA AAA_lid_14 HTH_8 GAF_3 His-Kinase His-Kinase DUF3485

JABURJ010000111.1:211-3246

Kangiella sp.

? GrpE RelE HTH_3 TACO1_YebC_N Transcrip_reg

JACXUJ010000057.1:7044-14520
Campylobacterales bacterium

Cas_Cas4 Cas3_HD ResIII Cas3-like_C_2 Cas_Cas5d Cas_Cas7 Cas_TM1802 Cas_Cas6_C MqsA_antitoxin RelE

JADCRD010000002.1:370260-385996
Lysobacterales bacterium

Bax1-I PseudoU_synth_2 Endonuclease_1 PPC LTD AceK_kinase AceK_regulatory HKD ? ? RelE HTH_3 Iso_dh ? 2OG-FeII_Oxy_3 Methyltransf_11 adh_short Hydrolase_4

JADFZT010000001.1:66630-80751

Magnetococcales bacterium

ExbD ? AICARFT_IMPCHas MGS Methyltr_RsmB-F RsmF-B_ferredox AI-2E_transport YkuD Bax1-I RelE MqsA_antitoxin Reprolysin_4 Big SLH SLH DHDPS Cytochrom_C Cytochrom_C SmpB Ppx-GppA

JADFYZ010000167.1:3085-4306

Magnetococcales bacterium

HTH_3 RelE FGS Transposase_31 DUF4351 ?

WGMY01000040.1:25467-31011
Alphaproteobacteria bacterium

ParG ? DUF6876 RP-C RP-C_C HTH_17 SNase HTH_3 RelE ?

JACOUS010000023.1:180174-18505
Ignavibacteriales bacterium

FRG PIN DUF4395 RelE HTH_3 Discoidin CHB_HEX_C_1 Discoidin Alpha_L_fucos Fucosidase_C Alpha_L_fucos Discoidin DUF4838 Glyco_hydro38C2 Glyco_hydro_38C Alpha-mann_mid Glyco_hydro_38N Discoidin Fn3_assoc Glyco_hydro_20 Glyco_hydro_20b SSF

JACOVE010000055.1:157-4876
Proteobacteria bacterium

HTH_3 RelE CHASE4 His-Kinase His-Kinase Receiver Receiver SpoIIE ?

JACPVQ010000115.1:8757-24257
Candidatus Solibacter usitatus

CarboxypepD_reg Plug TonB_dep_Rec_b-barrel Citrate_synt SocA_Panacea ? PhdYeFM_antitox PIN HTH_3 RelE DUF1501 PSCyt1 PSCyt2 PSD1 T4_deiodinase AhpC-TSA RoxA-like_Cyt-c Meth_synt_2 PD40 PD40 PD40 OpuAC PD40

JACQAS010000068.1:6889-9565
Chloroflexi bacterium

Mur_ligase_C LON_substr_bdg AAA Lon_lid Lon_C ? Peripla_BP_6 BPD_transp_2 BPD_transp_2 ? RelE HTH_3 AhpC-TSA

JACQDG010000011.1:5968-10390
Acidobacteria bacterium

PD40 PD40 Peptidase_S9 CarboxypepD_reg Plug TonB_dep_Rec_b-barrel MqsA_antitoxin RelE Acetyltransf_10 TPRs TPRs Ribonucleas_3_3

JACQOE010000042.1:1-8073
Proteobacteria bacterium

HUTI_composite_bact MFS_1 Phage_integrase RelE HTH_3 ? HTH_17 ? DUF6371 AAA_25 ? dnstrm_HI1420

JACQQL010000257.1:1-3932
Nitrospinota bacterium

HTH_31 RelE HTH_31 EamA EamA GATase IGPD Transketolase_C

JACQVH010000094.1:8719-14522
Candidatus Hydrogenedentota bacterium

Eco57I DNA_processg_A Calcineurin Calcineurin ? PsbP_2 RelE HTH_3 HEPN_AbiU2 Pect-lyase_RHGA_epim DUF433 Anticodon_1 tRNA-synt_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 ? MazE_antitoxin

JACRCV010000051.1:83243-104409
Deltaproteobacteria bacterium

Flagellin_N Flagellin_C FlbT FlaF MCPsignal ? Prim-Pol PriCT_1 HTH_17 RelE HTH_3 DUF6798 FGS STY-kinase ? Synaptojanin DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 NHL Vgb_lyase

JACRPZ010000005.1:51737-7844
Nitrospirae bacterium

HTH_8 AAA_lid_14 NtrC-AAA Receiver HTH_3 RelE ? ? TauE Alph_Pro_TM Usp Usp Usp DUF202

JACRLM010000099.1:24690-36442
Nitrospinota bacterium

PIN Plug TonB_dep_Rec_b-barrel adh_short_C2 Amino_oxidase Amino_oxidase PP-binding Ketoacyl-synt_C ketoacyl-synt RelE HTH_31 AMMECR1 NUDIX Aminotran_1_2

JADJCM010000002.1:400983-414254
Lysobacterales bacterium

DUF1365 Amino_oxidase FA_desaturase ? Uma2 ? HTH_3 RelE ? DUF4279 DEDD_Tnp_IS110 ? Alpha-amylase SusG_C SSF MFS_1 MFS_1

JADJIW010000147.1:73120-89401
Deltaproteobacteria bacterium

AAA_lid_14 NtrC-AAA NtrC-AAA GAF_2 ? ? ? MATCAP Acyltransferase RelE MqsA_antitoxin ? ? ? Cu-binding_MopE Cu-binding_MopE ? FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3

JADJYJ010000003.1:363629-383619
Candidatus Amarolinea dominans

ABC_membrane ABC_tran ? E1_dh Transketolase_C Transket_pyr 2-oxoacid_dh E3_binding Biotin_lipoyl Pyr_redox_dim Pyr_redox_2 HTH_3 RelE ? AAA_15 ABC-ATPase GAF_3 PAS His-Kinase His-Kinase ? FAD_binding_5 CO_deh_flav_C PBP ?

RBMQ01000010.1:47010-62317
Proteobacteria bacterium

Peptidase_M20 GATase_7 Asn_synthase OTCace_N OTCace SSF CbiM RDD HTH_3 RelE ? ? ? OCRE LptF_LptG LptF_LptG Peptidase_M17 Peptidase_M17_N

JAESUG010000108.1:178-3371
Robiginitomaculum sp.

Isochorismatase Pirin RelE MqsA_antitoxin peroxidase

JAESRV010000038.1:15735-24240
Amylibacter sp.

HTH_17 ? ? MarR_2 HTH_31 DUF2043 HTH_3 Gp49 RelE HTH_3 Zn_ribbon_recom Recombinase TraG-D_C ? ? MatE MatE

VGOC01000009.1:33986-51122
Chloroflexi bacterium

Usp Calcineurin Peripla_BP_3 LacI ? ? ? HTH_3 RelE TMEM164 DHH DHHA1 RecJ_OB OCD_Mu_crystall ? ABC_tran ABC_tran BCA_ABC_TP_C BPD_transp_2

JAFGEK010000026.1:1-6399
Deltaproteobacteria bacterium

? HicA_toxin UPF0150 HTH_3 RelE HTH_47 HTH_3 BrnT_toxin BrnA_antitoxin MqsA_antitoxin DUF4258 ? Fer4_7 Fer4_16 Fer4 Fer4_5 Fer4_5

JAFLDH010000274.1:1-6682
Burkholderiales bacterium

Beta-lactamase ? Ni_hydr_CYTB Hydrolase ? ? HTH_3 RelE Dus DDAH_eukar

JAGGAC010000041.1:17534-24767
Bacillota bacterium

Phage_P2_GpU GPW_gp25 Baseplate_J Tail_P2_I GDPD Phage_tail_fib ? HTH_3 RelE ? Amidase_2 CarboxypepD_reg

JAGBIY010000008.1:15680-31423
Treponema sp.

VCA0040-like VCA0040-like Receiver His_kinase His-Kinase Hydrolase MCPsignal SHMT SHMT HTH_3 RelE RtcB RtcB TPRs Semialdhyde_dh Semialdhyde_dhC_1 VWA ? Ribosomal_S21 NAGPA

JAGOBA010000005.1:162882-178927
Alphaproteobacteria bacterium

MatE MatE PEPCK_ATP AAA_33 ? DUF2971 Acyl_transf_3 ? RelE HTH_3 DNA_pol_A DNA_pol_A_exo1 5_3_exonuc 5_3_exonuc_N zf-CHCC His_Phos_1 Lactamase_B DUF4440 HTH_1 LysR_substrate

JAGOLX010000065.1:17303-28753
Phycisphaerae bacterium

? ? fn3 fn3 fn3 Beta_helix MR_MLE_C MR_MLE_N GFO_IDH_MocA_C2 GFO_IDH_MocA_C2 GFO_IDH_MocA ? HTH_3 RelE

JAGPIV010000038.1:10150-12892
Synergistaceae bacterium

tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 S4 PseudoU_synth_2 Fic-like_C LRRs ? RelE MqsA_antitoxin ? NFACT-R_1 NFACT_N NFACT_N HD_5 SBP_bac_11 PBP_like_2 BPD_transp_1

JAFWTP010000098.1:20995-32394
Neisseriaceae bacterium

cNMPBD HTH_Crp_2 DUF4064 Peptidase_M48 ? RsmF-B_ferredox Methyltr_RsmB-F YbjN YbjN Glutaredoxin RelE HTH_3 DCD Iron_traffic Bax1-I HRDC RQC RecQ_Zn_bind Helicase_C DEAD Adhesin_E HAP1_N

JAFECP010000005.1:549-18167
Proteobacteria bacterium

VIT1 DUF6496 PAS PAS PAS His-Kinase His-Kinase 2OG-FeII_Oxy_3 HNH Fer2 Reductase_C Pyr_redox_2 RelE HTH_3 TsaE APH PDDEXK_1 UvrD-helicase UvrD_C PDDEXK_1 Thioredoxin Hexapep Hexapep

JAFECP010000035.1:3022-15143
Proteobacteria bacterium

DUF979 DUF969 ? Cyt_bd_oxida_I Cyt_bd_oxida_II YbgT_YccB HTH_31 RelE FmdA_AmdA FmdA_AmdA HpcH_HpaI Abhydrolase_1 CinA NIPSNAP DUF1579 WCX WYL HTH_11

JAFECR010000030.1:54829-64192
Proteobacteria bacterium

T4BSS_DotI_IcmL Metal_hydrol GTP_EFTU EF-G_D2 RF3_C Na_sulph_symp ? ? MqsA_antitoxin RelE ? ? ?

JABGPE010000123.1:27953-39250
Gammaproteobacteria bacterium

? ? AAA PriCT_2 ? ? Phage_integrase Phage_int_SAM_5 RelE MqsA_antitoxin Phage_integrase LysR_substrate HTH_1 Methyltransf_31 CN_hydrolase GFA Thioredoxin_2 Thioredoxin_2

JABIIR010000094.1:144-7142
Rhodospirillaceae bacterium

RelE HTH_3 Meth_synt_2 DAO SO_alpha_A3 Pyr_redox_2 Fer2_4 Glycos_transf_2 DAO

JABJLO010000107.1:12817-21398
Thiotrichales bacterium

Guanylate_kin RNA_pol_Rpb6 ACT_4 RelA_AH_RIS TGS RelA_SpoT HD_4 Ribonuc_L-PSP RecG_dom3_C Helicase_C DEAD RecG_wedge Y1_Tnp Sigma70_r4 MqsA_antitoxin RelE Porin_4

JAHIPS010000013.1:490-7823
Candidatus Omnitrophota bacterium

MazE_antitoxin HTH_3 RelE ? RF-1 ? ? NYN ? FeoB_N

JAGWVX010000011.1:10192-12185
Alphaproteobacteria bacterium

ABC_tran ? His-Kinase His-Kinase HerA-FtsK Arm-DNA-bind_3 Phage_int_SAM_3 Phage_integrase RelE MqsA_antitoxin ? Nmad3 SHOCT-like_DUF2089-C ? SHOCT-like_DUF2089-C DUF2089 AAA_26

PCCX01000061.1:141920-148331
Nevskiales bacterium

Lipase_GDSL_2 TetR_N ? FMO-like NAD_binding_8 ? ? HTH_3 RelE

JAFANB010000121.1:992-8795
Burkholderiales bacterium

? ? ? PriCT_2 Toprim_3 DUF3631 Phage_AlpA ? HTH_3 RelE Arm-DNA-bind_3 Phage_int_M Phage_integrase Gln-synt_C

JAFATJ010000402.1:1-3298
Acidobacteriaceae bacterium

? RelE HTH_3 ? PepX_C Peptidase_S15

JAHBYG010000019.1:38024-50570
Ferrovibrio sp.

Ribosomal_L21p Ribosomal_L27 MMR_HSR1 GTP1_OBG ? Aldedh CTP_transf_like ? RelE Methyltransf_12 ADH_N ADH_zinc_N IF3_N IF3_C TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon ? NYN ?

JAIBAP010000041.1:22288-50285
Caldilineales bacterium

GATase_2 Glu_syn_central Glu_synthase GXGXG MFS_1 Aldo_ket_red SdrD_B GEVED DUF7507 SdrD_B DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 DUF7507 SdrD_B SdrD_B SpaA Big TED Amino_oxidase HAMP His-Kinase His-Kinase MTS Thump_like RelE HTH_3 ? TumE CinA CarboxypepD_reg Peptidase_S8 PcrA_UvrD_tudor UvrD_C UvrD-helicase RsgA_GTPase RsgA_N

JAIEOV010000074.1:25256-31137
Reyranella sp.

? ? Rrf2 Acetylase ECH_1 PaaA_PaaC HTH_3 RelE

JAIEOV010000135.1:142-6023
Reyranella sp.

RelE HTH_3 PaaA_PaaC ? ? Rrf2 ? ?

JAIOHO010000752.1:1-7485
Anaerolineae bacterium

? RelE HTH_3 DUF4145 REase ResIII Helicase_C EcoEI_R_C HsdM_N N6_Mtase Methylase_S

JAIQFN010000010.1:111812-121608
Terriglobia bacterium

Bac_transf CoA_binding_3 Pribosyltran Cytochrome_P460 MacB_PCD FtsX MacB_PCD FtsX PadR GDE_C GDE_N MqsA_antitoxin RelE ? ?

JAGQKL010000567.1:1-2929
Candidatus Omnitrophota bacterium

CTP_transf_1 TerC PGPGW MqsA_antitoxin RelE

JAGRLI010000178.1:297-1867
Rhizobiaceae bacterium

? HTH_3 RelE ?

JAGREC010000317.1:1-3427
Rickettsiales bacterium

Peptidase_M10_C Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll P_proprotein Peptidase_S8 CMD HTH_3 RelE

JACKIM010000002.1:1869837-1886618
Geminicoccaceae bacterium

Cytochrom_C CTP_transf_3 PDT TOBE_2 OB_MalK MIT_C ABC_tran Gly-zipper_YMGG Rick_17kDa_Anti YmgD HTH_3 RelE ? ? YchJ_M-like ACR_tran HlyD_3 SNF2-rel_dom Helicase_C ?

JAJFBR010000092.1:1-5322
Gammaproteobacteria bacterium

DNA_pol3_delta2 HTH_3 RelE Thymidylate_kin AAA_14 DUF4143 UPF0242 YceG ?

JADLDZ010000155.1:6851-16612
Nitrospirales bacterium

? 5_3_exonuc 5_3_exonuc_N ROK GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C Peptidase_S26 Lum_binding Lum_binding VapB_antitoxin RelE HTH_3 Lactamase_B_2

JADLDO010000028.1:811360-825111
Caldilineaceae bacterium

? HpcH_HpaI PIN ? Ldh_2 Ldh_2 MR_MLE_C MR_MLE_N adh_short_C2 RelE MqsA_antitoxin Peptidase_S15 LysM ? Aldo_ket_red Amidohydro_3 Amidohydro_1 EF_0837-like_N AAA_33 TrkA_C TrkA_N Na_H_Exchanger

JADLDH010000202.1:43192-54234
Bryobacterales bacterium

ParE_toxin Glycos_transf_2 Epimerase Glyco_transf_9 TPRs TPRs TPRs TPRs TPRs GHMP_kinases_N GHMP_kinases_C NTP_transferase SIS_2 Hydrolase_like RelE MqsA_antitoxin DUF5615 DUF433 YoeB_toxin PhdYeFM_antitox HTH_37 GFO_IDH_MocA GFO_IDH_MocA_C3

JADLAX010000077.1:26090-46998
Anaerolineae bacterium

ABC2_membrane_3 ABC_tran ABC2_membrane_3 NUDIX FHA FHA HTH_3 RelE Asp-Al_Ex TrkA_C TrkA_C Asp-Al_Ex AAA_16 STY-kinase AAA_16 TPR_MalT TPRs TPRs ? polyprenyl_synt BTAD NACHT TPRs TPRs TPRs ?

JADZBO010000161.1:8174-17296
Burkholderiaceae bacterium

Porin_4 TumE HVO_A0114 HTH_31 HicB_lk_antitox HicA_toxin GGDEF RelE HTH_3 ? PRiA4_ORF3 DUF2188 ? ARG

JADYUI010000001.1:63395-75832
Roseburia hominis

MMR_HSR1 YchF-GTPase_C Phage_int_SAM_3 Phage_integrase Tn916-Xis ? ? Phage_integrase HTH_3 RelE NikA-like HNH ? DUF5090 Bac_DNA_binding dCache_1 HAMP MCPsignal TPK_catalytic TPK_B1_binding

JAKLQI010000272.1:1066-7519
Syntrophaceae bacterium

Radical_SAM_C Radical_SAM TrbI_Ftype ? AAA_14 DUF4143 ParD_like ParE-like_toxin SLFN_AlbA_2 HATPase_c_4 HTH_24 RelE HTH_3 ?

JAKSBH010000319.1:1-6612
bacterium

? AAA_22 TPR_MalT GerE DUF4440 MFS_1 MFS_1 RelE HTH_3

JADFJB010000009.1:15582-30473
candidate division KSB1 bacterium

Shikimate_dh_N Shikimate_DH SDH_C Chorismate_synt PMSR HTH_3 AAA_19 UvrD_C UvrD_C ? ? RelE HTH_3 WbqC Amidohydro_3 HemN_C Radical_SAM Peptidase_S26 LepA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU DUF5658

JAGLFB010000472.1:709-1834
Gammaproteobacteria bacterium

HTH_3 RelE DUF3010

JALKGA010004467.1:90677-103373
Microbacteriaceae bacterium K1510

SET Sugar_tr Ldh_1_C Ldh_1_N DUF5996 ? DPBB_1 Gly-zipper_YMGG RelE HTH_31 DUF4019 Gp49 HTH_3 Fumerase Fumerase_C Amidohydro_2 TctC

WRGB01000067.1:1-2437
Treponema sp.

Ribosomal_S9 Ribosomal_L13 RelE ? ?

WRKN01000029.1:3163-16992
Synergistaceae bacterium

N_methyl Aminotran_1_2 IGPD His_biosynth His_biosynth PRA-CH PRA-PH PALP Peptidase_M20 RelE MqsA_antitoxin SBP_bac_3 SBP_bac_3 PAS PAS His-Kinase His-Kinase Receiver Hpt tRNA_U5-meth_tr tRNA_U5-meth_tr TRAM

JAMDEG010000914.1:1-9358
Acidobacteria bacterium

HD_5 UDG MqsA_antitoxin RelE Xan_ur_permease ? SPASM Radical_SAM ? DUF302 CcmH Glyco_tranf_2_3

JAMPDU010000152.1:1-6812
Ruminococcus sp.

LytTR PAF-AH_p_II O-antigen_lig Flg_new Flg_new FecR RelE HTH_3 ?

JAGFIY010000122.1:51959-65264
Melioribacteraceae bacterium

YtxH LPG_synthase_TM Maf ? HTH_3 FG-GAP_3 Por_Secre_tail RelE HTH_3 PhoD DUF7800 ? DUF123 Epimerase Aminotran_1_2 ?

JAAENE010002868.1:111-3002
bacterium

MNS-STAND ? HTH_3 RelE

JAAEOU010001400.1:178-1194
bacterium

PEP-CTERM RelE MqsA_antitoxin

JAAEOZ010000281.1:1-3254
Gammaproteobacteria bacterium

HTH_18 RelE HTH_3 Methyltransf_23 Choline_kinase DHDPS

JAAETX010001207.1:1853-13703
Planctomycetota bacterium

T4SS-DNA_transf ? MobA_MobL VirC1 ? ? DUF3768 RelE HTH_3 AAA_15 OLD-like_TOPRIM ? ?

JALOQK010000002.1:438864-11533
Candidatus Omnitrophota bacterium

PrlF_antitoxin YafQ_toxin ORF6N RelE MqsA_antitoxin ? CxxCxxCC Receiver ? PCC ?

JAHXHH010000501.1:244-1566

gamma proteobacterium symbiont of Bathyaustriella thionipta

RHH_1 ParE_toxin RelE HTH_3

JAPLUD010000360.1:5274-18034
Verrucomicrobia bacterium

Sigma70_r2 Reg_prop DUF6786 AbiEi_2 AbiEii ? HTH_3 RelE FGS HATPase_c_4 SLFN_AlbA_2 Helicase_C N_methyl ANKs Sigma70_ECF

JAPLRG010000347.1:289-9098
Gammaproteobacteria bacterium

MqsA_antitoxin RelE ChlI Mg_chelatase Mg_chelatase_C BMFP fn3 Thaumatin fn3 fn3 PALP Cys_Met_Meta_PP ? SH3_3

JAPLOI010000006.1:22068-37599
Alphaproteobacteria bacterium

DapB_N DapB_C ? ParE_toxin RelB PaaA2 Gly_transporter Gly_transporter Methyltransf_11 HTH_3 RelE DnaJ DnaJ_C HSP70 GrpE dCache_1_like PAS His-Kinase His-Kinase Receiver Hpt Receiver EAL HrcA Peptidase_M20

JAPPAT010000147.1:581-2243
Gammaproteobacteria bacterium

Sulfotransfer_1 KwaB RelE FRG

JAQTYN010000033.1:20552-37367
Gallionella sp.

DUF4139 PTPS HKD SNF2-rel_dom SNF2-rel_dom Helicase_C MmeI_Mtase Eco57I ? DNA_processg_A DUF1016_N YhcG_C RelE HTH_3 PG_binding_1 YkuD Peptidase_M23 ? DUF2934 CsbD Ferritin

JAQTXN010000133.1:194-7030
Methylococcales bacterium

Rv2743c ? RelE MqsA_antitoxin WCX WYL HTH_11 Lactamase_B Competence DUF4131 DUF2062 ?

JAQVCS010000001.1:248722-259371
Desulfovibrionaceae bacterium

TsaD ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD BOF ATP_bind_3 TtuA_LIM_N HemN_C Radical_SAM tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c Arm-DNA-bind_3 Phage_int_M Phage_integrase RelE MqsA_antitoxin ? ?

JAPPMH010000118.1:35329-52048
Gammaproteobacteria bacterium

MmgE_PrpD_N MmgE_PrpD_C ABC_tran FecCD Peripla_BP_2 ? GMC_oxred_C GMC_oxred_N HTH_3 RelE MatE MatE Lactamase_B ? Gln-synt_C Gln-synt_N Gln-synt_C DAO MS_channel_3rd MS_channel_2nd MS_channel_1st MS_channel_1st_1

JAPPCO010000157.1:1-3779
Gemmatimonadota bacterium

IstB_IS21 HTH_17 Phage_integrase MqsA_antitoxin RelE Glyco_hydro_108

JAOTST010000195.1:1-3691
Chromatiales bacterium

Cpn60_TCP1 Usp DUF5610 HTH_3 RelE

JAUSID010000251.1:47472-65840
Opitutaceae bacterium

DDE_Tnp_1_4 ? Sigma70_r2 Sigma70_r4_2 Rce1-like CarboxypepD_reg Plug TonB_dep_Rec_b-barrel GFO_IDH_MocA GFO_IDH_MocA_C3 HTH_3 RelE Sulfatase SGSH_C ? DMFA2_C ? ? CN_hydrolase Glyco_hydro_100 DUF608 S6PP

JAUYPB010000050.1:291-4418
bacterium

HTH_3 RelE Phage_integrase Phage_integrase Phage_integrase Phage_int_SAM_1

JALYQJ010000224.1:346-10681
Acidobacteriota bacterium

Phage_int_SAM_4 Phage_integrase MMR_HSR1 MMR_HSR1 KH_dom-like RelE ? MauE AhpC-TSA AhpC-TSA TGT YajC Sec_GG SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C TonB_C

JAVHLK010000003.1:50818-61518
Desulfomonilaceae bacterium

RdgC ? SpoIIE PAS SBP_bac_3 STAS His-Kinase ? HTH_3 RelE HTH_3 RelE ? BrnT_toxin CopG_antitoxin ? ?

JAIROV010000223.1:148-4408
Rickettsiales bacterium

? CbiG-like_linker YoeB_toxin ? ? ? MqsA_antitoxin RelE ? Peptidase_S24 HTH_3 ?

JAISQV010000171.1:8191-7794
Spirochaetaceae bacterium

URO-D RelE HTH_31 PhdYeFM_antitox ABC_tran BCA_ABC_TP_C ABC_tran BPD_transp_2 BPD_transp_2 Peripla_BP_6

JAITRY010000314.1:1-5110
Treponema sp.

HTH_22 PAS Nitroreductase HTH_3 RelE UPF0175 AMP-binding PseudoU_synth_2

JARLJD010000064.1:17778-27626
Alphaproteobacteria bacterium

? DNA_binding_1 HD_4 PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV LptE tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 tRNA-synt_1 Anticodon_1 HTH_3 RelE Glyco_transf_9 TPRs TPRs TPRs TPRs TPRs

JAVVEJ010000036.1:14594-23123
Mariprofundaceae bacterium

RimM PRC_RimM tRNA_m1G_MT Coprogen_oxidas Cation_ATPase_C HAD Cation_ATPase E1-E2_ATPase Cation_ATPase_N ? HNH ? RelE HTH_3 ? DinB ?

JANRJL010000039.1:13008-10441
Rhodospirillaceae bacterium

DUF1800 DUF1501 HTH_3 Arm-DNA-bind_3 Phage_int_SAM_3 Phage_integrase RelE HTH_3 ? HTH_17 ? ? Eco57I TaqI_C XhoI

JANRJB010000410.1:14324-23073
Rhodospirillaceae bacterium

DUF1838 PQQ_2 MC_hydratase Polysacc_deac_1 adh_short_C2 MFS_1 HTH_3 RelE TPRs TPR_Slam ?

JARYGU010000030.1:875-11557
Shinella sp.

LysE His_Phos_1 DUF4037 Phage_AlpA RelE HTH_3 ? ? SGL TctA ? HMG-CoA_red

JAXHUR010000023.1:36033-42802
Thermodesulfobacteriota bacterium

GST_N_2 GST_C_2 Lactamase_B_2 PagL Aldo_ket_red Fer4_9 TetR_C_13 DDE_Tnp_IS66_C DDE_Tnp_IS66 HTH_3 RelE

JAWXCE010000026.1:46692-58592
Hyphomicrobiales bacterium

? HlyD_3 ? ? Beta_helix Pect-lyase_RHGA_epim DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase MqsA_antitoxin RelE ? PEMT ? Helicase_C DEAD Fe-S_biosyn FeS_assembly_P Aminotran_5

JAXDMS010000943.1:1-5810
Acetobacteraceae bacterium

Arginosuc_synth Glyco_trans_1_4 Glyco_trans_4_4 Glyco_trans_1_4 RelE HTH_3 Hydrolase 4HBT ROK

JAXJZR010000631.1:1597-2507
Desulfovibrio fairfieldensis

AMP-binding_C_2 AMP-binding HTH_3 RelE

SBEN01002822.1:591-1006
bacterium

Fer4_22 RelE

VXRD01000022.1:107938-120370
Gemmatimonadetes bacterium

HsdM_N N6_Mtase Methylase_S Methylase_S ? SMODS-polBeta Patatin HTH_3 Peptidase_M78 RelE HTH_3 UPF0150 ? SDH_protease ? ? Uma2

VYEH01000437.1:590-3099
Proteobacteria bacterium

ParB_N RelE HTH_31 PepSY_TM

JAACTN010000085.1:1-7664
bacterium

Peptidase_M29 HTH_3 RelE adh_short_C2 ? PIN ADH_N ADH_zinc_N HATPase_c_3 DNA_mis_repair MutL_C Hydrolase PDZ_6 CtpB_N-like

WWBY01000560.1:444-5695
Candidatus Aminicenantes bacterium

? ? gag-asp_proteas ? Lactamase_B ? HTH_3 RelE FGS

VEPA01000062.1:1078-14406
Betaproteobacteria bacterium

? Toprim DUF927 ? ? HTH_17 ? HTH_3 RelE Arm-DNA-bind_3 Phage_int_M Phage_integrase ZapD CoaE SPOR AdoMet_dc NAD_binding_8 Spermine_synth

JAAZOK010000150.1:18-9815
Chloroflexota bacterium

DnaJ_CXXCXGXG Hypoth_Ymh HTH_3 RelE PD40 PD40 PD40 PD40 PD40 AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C3 BPD_transp_2 ABC_tran ABC_tran Peripla_BP_4 LacI Peripla_BP_1

JABMPA010000361.1:228-3108
Rhodospirillales bacterium

HTH_31 RelE AMP-binding AMP-binding_C adh_short_C2

JABSOJ010000541.1:181-2431
Alteromonadaceae bacterium

Transposase_20 HTH_3 RelE Phage_integrase

JABSOW010000017.1:40713-54992
Colwellia sp.

YagK_YfjJ_C YagK_YfjJ_C YagK_YfjJ_C LRAT Phage_AlpA Bac_DNA_binding Phage_integrase RelE MqsA_antitoxin Plug TonB_dep_Rec_b-barrel Sulfatase SGSH_C Sulfatase DUF6429 Paired_CXXCH_1 Cytochrom_C552 HEATs rhaM

JABSRB010000034.1:32090-48263
Oceanospirillaceae bacterium

MazG-like RE_AspBHI_N Mrr_cat DNA_methylase HATPase_c_3 HEF_HK His-Kinase Response_reg_2 HTH_3 HigB-like_toxin RelE HTH_3 HTH_3 ? ? HipA_C Couple_hipA HTH_3 EAL GGDEF HAMP CHASE5

JABWBV010000072.1:4751-17167
Anaerolineales bacterium

MFS_1 MFS_1 Aldo_ket_red NHL NHL DUF5128 DUF5128 DUF5128 Methyltransf_25 DAO ? HTH_3 RelE MM_CoA_mutase HTH_17 Glycos_transf_2 His-Kinase His-Kinase GAF_2 GAF GAF_2

JACADU010000351.1:4403-17975
Bryobacteraceae bacterium

Hemerythrin dCMP_cyt_deam_1 Aminotran_1_2 GntR ThiG SOR_SNZ SNO LMWPc HTH_3 RelE HicB-like_2 HTH_24 dCMP_cyt_deam_1 RibD_C SRP54_N SRP54 Hepar_II_III_C Hepar_II_III_N ? 3keto-disac_hyd ?

LWQU01000161.1:3734-10685
Magnetospirillum moscoviense

dnstrm_HI1420 ? RelE ? GST_N_2 GST_C LeuA_dimer IPMS_D2 HMGL-like OMP_b-brl ? ? His-Kinase PAS

MERG01000053.1:1027-9545

Betaproteobacteria bacterium RIFCSPLOWO2_12_FULL_62_13

Gp49 dnstrm_HI1420 ? HTH_3 RelE ? ? ECH_1 TctC PRISE Epimerase FAD_binding_2 FAD_binding_2

MFNF01000007.1:587-9075

Candidatus Lambdaproteobacteria bacterium RIFOXYD2_FULL_56_26

? AntA AntA NikA-like MFS_1 TraG_N ? HTH_3 RelE Phage_integrase Phage_int_SAM_5

MGYB01000009.1:13677-12023

Gammaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_41_15

Ribosomal_L28 Ribosomal_L33 SBP_bac_3 HTH_3 RelE Peptidase_M23 Csd3_N2 LysM_OapA AnmK ? DUF4143 AAA_14 CDP-OH_P_transf AmiA-like tRNA-synt_1b S4 PALP

MHAE01000077.1:462-4797

Ignavibacteria bacterium RBG_13_36_8

? HTH_3 Zincin_1 ? RelE

MHBD01000056.1:15619-7535

Legionellales bacterium RIFCSPHIGHO2_12_FULL_42_9

HTH_3 RelE MPTase-PolyVal RVT_1 GIIM DUF155 RIH_assoc ? GDC-P GcvP2_C GDC-P YCII

MTEJ01000581.1:323-4594
Thiothrix lacustris

HTH_3 RelE Mrr_N Mrr_cat SBP_bac_3 Transgly_assoc Phytase-like

MTEJ01000446.1:231-5709
Thiothrix lacustris

KfrA_N CENP-F_leu_zip ? ? Rep3_N Rep3_C HTH_73 HTH_3 RelE

NBMI01000023.1:24505-35942

Candidatus Parabeggiatoa sp. nov. 2

HicA_toxin ? ? ? Transgly HTH_3 RelE ? DUF2530 ? MinE ParA-Soj-NTPase MinC_N MinC_C ? UPF0150

NVTL01000019.1:12021-27031
Rhodospirillaceae bacterium

N6_N4_Mtase HEPN_Swt1 HEPN_AbiU2 SNF2-rel_dom Helicase_C NikA-like TraI_middle Relaxase MqsA_antitoxin RelE Imm_superinfect ? Phage_integrase Phage_int_M PFK Receiver LysE Methyltransf_11

NVVB01000125.1:1-7323
Gammaproteobacteria bacterium

DUF4019 SMI1_KNR4 HTH_3 PhdYeFM_antitox PhdYeFM_antitox ? ? Acetylase ? RelE PIN PIN

NVVW01000013.1:45002-63041
Methylophaga sp.

CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS CPSase_sm_chain GATase CN_hydrolase ALAD SDH_C Shikimate_DH Shikimate_dh_N ? HTH_3 RelE ? PP-binding His-Kinase DUF4325 Adenine_glyco ? DNA_pol3_tau_5 DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 YbaB_DNA_bd

PHBU01000016.1:60023-77097
Chloroflexi bacterium HGW-Chloroflexi-6

PfkB ? Band-7 Transgly HI0933_like ? HTH_3 RelE Thiolase_C_1 Thiolase_N HNH_5 DUF1731 Epimerase ? Rce1-like SYY_C-terminal tRNA-synt_1b SKI DHQ_synthase DHQS_C

PHEN01000007.1:6557-23831
Alphaproteobacteria bacterium HGW-Alphaproteobacteria-12

MutS_V MutS_III MutS_II MutS_I malic Malic_M PTA_PTB Acetyltransf_5 MFS_1 Receiver His-Kinase His-Kinase PAS Mg_chelatase_C Mg_chelatase ChlI ? RelE MqsA_antitoxin Peptidase_M19 ? GSH-S_ATP GSH-S_N MarC TPRs Transglut_core2

QAYZ01000012.1:20043-33359
Spartobacteria bacterium LR76

T2SSE Pan_kinase Cys_rich_CPXG 1-cysPrx_C AhpC-TSA CMD HTH_3 RelE RsfS CTP_transf_like MarR HlyD_D23 ACR_tran ? T2SSE Glyco_hydro_3

CP064970.1:2034530-2042432
Alphaproteobacteria bacterium

? ? DUF2303 ? zf-C4_ClpX ? POU2AF2 RelE HTH_3 HTH_3 HTH_31 ? DUF4145 VRR_NUC HTH_31

QFXH01000003.1:140434-156832
Candidatus Nitrotoga sp. MKT

ABC2_membrane Phage_integrase ? Sensor SMEK_N MC6 AAA_23 DUF2326 RelE HTH_3 Zn_peptidase MFS_1 ? EamA EamA His-Kinase His-Kinase HAMP CusS Trans_reg_C Receiver

QEVO01000007.1:170-7295
Proteobacteria bacterium

? SpoU_sub_bind SpoU_methylase RelE HTH_3 Lactamase_B HAGH_C SDF SelR ? DUF885 GFA

RFLB01001700.1:127-1667
Bacteroidetes bacterium

? ? RelE HTH_3

FOSY01000005.1:46189-66452
Lachnospiraceae bacterium KH1T2

Receiver GerE DUF6057 HisKA_3 His-Kinase Peripla_BP_4 GPDPase_memb GDPD GGDEF EAL GGDEF Resolvase HTH_3 RelE Nitroreductase Nitroreductase Zn_ribbon_15 AAA ? ? AAA ?

FUWR01000003.1:134112-147521

Trichlorobacter thiogenes

Pantoate_transf ? GGGtGRT ? DUF190 CRCB HTH_3 RelE PAP2 ? Voltage_CLC HTH_1 LysR_substrate Cons_hypoth698 GAF NTase-conflict MCPsignal ?

OKRH01000112.1:2916-14987
Candidatus Sulfopaludibacter sp. SbA6

ABC2_membrane Amidohydro_1 ABC_tran DrrA1-3_C DUF4177 ? ? ? HTH_3 RelE AMP-binding_C_2 AMP-binding ACT_8 HTH_3 TIR AP-ATPase APAF1_C Beta-prop_WDR3_1st Beta-prop_WDR3_1st WD40_CDC20-Fz Beta-prop_EML_2 ThiS Oxidored_q4 Complex1_30kDa

OMOG01000237.1:14705-14561
Candidatus Sulfopaludibacter sp. SbA4

MacB_PCD MacB_PCD FtsX MacB_PCD FtsX Xre_MbcA_ParS_C MqsA_antitoxin RelE HTH_37 ? ? STAS CarboxypepD_reg HZS_alpha Sel1 Sel1 Sel1 Sel1 Sel1 Amidohydro_3 Amidohydro_3 Urease_alpha

UGOD01000003.1:8911-17687

Legionella busanensis

? HTH_3 CinA NAD_synthase ? Hydrolase_4 HTH_3 RelE ? DUF5710 ? ? Peptidase_S24 ? DUF3800

SCUW01001065.1:65-4674
bacterium

UPF0060 RelE HTH_3 Glyoxalase FMN_dh FMN_dh LemA Peptidase_M48

SMCN01000047.1:4170-11662
Methylomonas methanica

HTH_Tnp_1 Amino_oxidase EutB EutC HTH_Tnp_1 ? rve rve_2 RelE HTH_3 rve_3 rve

SZXW01000010.1:913297-920795
Chlorobium sp.

HD-CE MqsA_antitoxin Mu-transpos_C_2 rve IstB_IS21 RelE ? ? RelE ?

VAZY01000163.1:6157-12728
Alphaproteobacteria bacterium

Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 CoA_transf_3 CoA_transf_3 DUF6640 HTH_31 RelE ? DctM DctP FMN_bind_2 MaoC_dehydratas CopD Cytochrome_CBB3 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1

NZ_AEWB02000012.1:23567-33000
Aggregatibacter aphrophilus ATCC 33389

Ribosomal_L11_N Ribosomal_L11 HTH_3 T6SS_HCP SecE NusG KOW Ribosomal_L10 Ribosomal_L1 DUF4455 ? ? HTH_24 AntA ? RelE ?

NZ_AP027077.1:346740-362004
Haemophilus influenzae

Beta_elim_lyase Trp_Tyr_perm ? ATPase_2 Peptidase_M23 LysM AntA Phage_ASH ? ? ? ? DUF927 RelE MqsA_antitoxin MutS_V MutS_III MutS_II MutS_I

NZ_CP032663.1:2656969-2670192
Proteus vulgaris

EIIC-GAT Acetate_kinase DUF412 HOK_GEF NUDIX HAD_2 LacI Peripla_BP_3 PTS_EIIA_2 PTS_IIB YfbU AAA_26 DRTGG PTA_PTB Ala_racemase_C Ala_racemase_N RelE GST_N_2 GST_C HTH_3

NZ_AGCA01000069.1:680-2295
Candidatus Regiella insecticola 5.15

RelE HTH_19 Tra_M

NZ_NZYJ01000081.1:1158-21178
Salinisphaera sp.

HTH_3 ? BPD_transp_1 ABC_tran TetR_C_13 TetR_N FA_desaturase Rubredoxin BPD_transp_1 ? RelE RecD_N AAA_19 SH3_13 UvrD_C_2 UvrD-helicase UvrD_C UvrD_C PDDEXK_1 Exonuc_V_gamma RecC_C FG-GAP ? ?

NZ_CP133110.1:387594-401916
Pseudomonas chlororaphis subsp. aurantiaca

FMN_red BPD_transp_1 ? ? Methyltransf_21 ? HTH_18 Arabinose_bd VRP3 VRP3 RelE ? Methyltransf_32 BPD_transp_1 HTH_3 Phage_base_V

NZ_AJXS01000105.1:10465-15347
Rhodanobacter sp. 115

? ? ? ? HTH_17 Arc HTH_3 RelE Arm-DNA-bind_3 Phage_integrase Exonuc_V_gamma

NZ_JH114304.1:386213-396390
Desulfovibrio sp. 6_1_46AFAA

HTH_31 RusA DUF6604 Terminase_6C ? ? HTH_3 Pesticin GTA_holin_3TM ? ? ? Phage_capsid_4 RelE ?

NZ_AFPM01000037.1:5764-18994
Oceanicola sp. S124

Histidinol_dh DUF2948 EPSP_synthase ? ? ? HTH_17 RelE HTH_3 ? ? ? HTH_23 ? HTH_3

NC_018020.1:3664251-3676651
Turneriella parva DSM 21527

? ? MazE_antitoxin PIN PhdYeFM_antitox YoeB_toxin HTH_3 ? ATP-synt_B DUF2764 ATP-synt_ab_N ATP-synt_ab_Xtn ATP-synt_ab ATP-synt_VA_C ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt_D RelE ?

NZ_KB902785.1:72634-89693
Bradyrhizobium sp. WSM2793

Dabb Chorismate_bind Anth_synt_I_N GATase ? RelE ? BPD_transp_1 adh_short HAD_2 SBP_bac_3 HTH_3 DUF6088 GATase_7 ? ?

NZ_KB892865.1:33702-54483
Uliginosibacterium gangwonense DSM 18521

? Ogr_Delta Ogr_Delta ? Phage_GPA Phage_integrase Phage_int_SAM_3 Arm-DNA-bind_2 HTH_31 RelE N_methyl N_methyl ART-PolyVal ? ? ? ? ?

NZ_DBGQ01000036.1:89584-105698
Spongiibacter tropicus

Biotin_carb_C CPSase_L_D2 Biotin_carb_N HTH_1 LysR_substrate Biotin_lipoyl PYC_OADA HMGL-like TonB_dep_Rec_b-barrel Plug TonB_dep_Rec_b-barrel ? RelE ? ? ? DUF927 HTH_39 HTH_17 ? MqsA_antitoxin

NZ_JHZK01000022.1:25272-42333
Afifella pfennigii DSM 17143

GCS2 ? Methyltrans_RNA PUA_4 tRNA_anti_2 Exonuc_VII_L Exonuc_VII_L RelE LutB_C Fer4_8 LUD_dom CCG CCG DctP GGDEF DctM LUD_dom PAS His-Kinase His-Kinase DctQ 4HBT HTH_3

NZ_AUKC01000061.1:847-10061
Butyrivibrio sp. NC3005

RelE HTH_3 DUF5643 DUF4179 Sigma70_r4_2 Sigma70_r2 Zn_Ribbon_1 Glycophorin_A Nup35_RRM_2 DUF1565 Beta_helix LytTR CxxCxxCC

NZ_RQYC01000001.1:239085-253376
Conchiformibius steedae

ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt_DE_N ATP-synt_DE ? ? LptE DNA_pol3_delta_C DNA_pol3_delta tRNA-synt_1b ATPase-cat_bd HMA E1-E2_ATPase Hydrolase HTH_3 RelE CM_2 PDT ACT ? HSP70 ATP-synt HSCB_C DnaJ

NZ_AZNY01000003.1:228974-240367
Pseudomonas sp. MOIL14HWK12:I1

HTH_3 HAD_2 Acetylase TetR_N TetR_C_13_2 GST_C_2 GST_N adh_short RelE DUF924 DJ-1_PfpI HTH_18 adh_short adh_short MFS_1 MFS_1 CMD HTH_18 Cupin_6 Esterase

NZ_BRLG01000006.1:176351-195916
Budvicia aquatica

MFS_1 HTH_31 RelE ? DUF3142 TetR_C_11 TetR_N DUF418 HGSNAT_cat DUF4056 ? ? ? UvrD-helicase AAA_15 OLD-like_TOPRIM ? GATase NAD_synthase GMP_synt_C

NZ_JBFWJW010000014.1:50010-10390
Hydrogenovibrio sp.

CsrA AA_kinase ACT ACT_9 RecA RecA_C ? RelE PIN HTH_3 Arm-DNA-bind_3 Phage_int_M Phage_integrase tRNA-synt_2c tRNA_SAD DHHA1 RecX_HTH1 RecX_HTH2 RecX_HTH3

NZ_CP053676.1:1004093-1016271
Ferrovum myxofaciens

RelB Polysacc_synt_3 Methyltransf_23 Glyco_trans_2_3 Glycos_transf_2 Mu-transpos_C_2 HTH_31 RelE ? HTH_32 RepB-RCR_reg AAA ? HTH_37 rve

NZ_BNAB01000017.1:2119-16384
Allgaiera indica

HTH_3 RHH_1 ? Peptidase_M41 AAA ? Oxidored_q3 Oxidored_q2 Proton_antipo_M Proton_antipo_N Proton_antipo_M Proton_antipo_M T5orf172 adh_short_C2 adh_short RelE ?

NZ_JQSG02000003.1:615506-631047
Acidihalobacter prosperus

FabA PPK2 PPK2 ThiC-associated ThiC_Rad_SAM HTH_3 PCMT Isochorismatase CN_hydrolase ? ? RelE ketoacyl-synt Ketoacyl-synt_C ? ? ? Glycos_transf_N Glycos_transf_1

NZ_BBMZ01000009.1:132057-150018
Pseudescherichia vulneris NBRC 102420

? DUF2526 DMT_YdcZ Acetylase ADH_N_2 ADH_zinc_N GntR FCD RelE HTH_3 TonB_dep_Rec_b-barrel Plug YNCE AA_permease Peptidase_M20 Acetylase SBP_bac_5 Bac_luciferase

NZ_JSEH01000003.1:67000-82243
Desulfovibrio sp. TomC

FmdE zf-Tbcl_FmdE MoCF_biosynth MarR Molybdop_Fe4S4 Molybdopterin Molydop_binding Fer4 Fer4_10 Fer4_11 Fer4_7 RelE ? Methyltransf_24 STAS GGDEF Fe-ADH ADH_Fe_C MFS_1 SLT HTH_3 FemAB_like

NZ_BJYZ01000035.1:37111-52216
Skermanella aerolata

LysE Rrf2 CitMHS NMT1_2 MqsA_antitoxin Cys_Met_Meta_PP F_bP_aldolase DUF411 BPD_transp_1 SATase_N Hexapep Hydrolase E1-E2_ATPase HMA ABC_tran ? TauD RelE

NZ_JBNNHP010000023.1:23451-42718
Vreelandella aquamarina

ELFV_dehydrog_N ELFV_dehydrog N6_N4_Mtase Suc_Fer-like Rnk_N GreA_GreB TonB_dep_Rec_b-barrel Plug DUF4391 Helicase_C SNF2-rel_dom AzlC AzlD HTH_3 RE_endonuc ResIII HATPase_c_4 SLFN_AlbA_2 RelE GGDEF

NZ_LAJY01000428.1:1284-4529
Elstera litoralis

? zf-dskA_traR DksA_N HTH_31 RelE FliX Rod-binding

NZ_LGVG01000042.1:19513-33033
Achromobacter spanius

Cloacin zf-C4_ClpX ? Glutaredoxin ParB_N HTH_ParB RelE HTH_3 ? Pyr_redox_2 BLH_phosphatase Lactamase_B YgaP-like_TM Pyr_redox_dim Pyr_redox_2 PAS NtrC-AAA AAA_lid_14 HTH_8 RMF DUF6900

NZ_CAAAIE010000005.1:262528-279906
Legionella parisiensis

? DUF4440 Transpeptidase DUF4156 ? HTH_7 Resolvase RelE MqsA_antitoxin ABC-ATPase OLD-like_TOPRIM Beta-lactamase ? DUF4158 DUF7624 Phage_integrase STY-kinase ANKs

NZ_CP012373.2:30728-45893
Beggiatoa leptomitoformis

Phage_coatGP8 ? DUF5455 Zot Zot ? FPP HTH_31 RelE Phage_integrase Tropomyosin Mrr_cat Baculo_PEP_C AAA_15 NTP_transf_3 ? MoaC YkuD

NZ_BAAAEQ010000002.1:844556-867190
Thalassospira tepidiphila

ABC_tran GntR FCD SBP_bac_3 BPD_transp_1 Aminotran_5 Aminotran_5 Amidohydro_1 Urease_alpha HTH_3 RelE UvrD-helicase UvrD-helicase UvrD_C ABC-ATPase OLD-like_TOPRIM Dabb FAD_binding_4 FAD-oxidase_C Fer4_8 ShlB POTRA_3 POTRA_2 DUF637 Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg Fil_haemagg TPS Imm40

NZ_CP011971.1:552794-568529
Steroidobacter denitrificans

DDE_Tnp_1 HTH_1 LysR_substrate Rieske Pyr_redox_2 Reductase_C ADH_N ADH_zinc_N HTH_37 HTH_3 ? Transcrip_reg TACO1_YebC_N DUF1840 HDOD Lactamase_B BLACT_WH ? RelE AAA_14 DUF4143 ABC_tran ABC_tran

NZ_JADGKV010000001.1:344796-355827
Pasteurella sp. 19428wF3_WM03

RelE HTH_3 NapD Fer4_7 Fer4_7 ? ? ? Fer4_7 Fer4_7 TAT_signal Molybdop_Fe4S4 Molybdopterin Molydop_binding BrnA_antitoxin DUF927 ? ? ? Arm-DNA-bind_3

NZ_MUET01000102.1:5978-5529
Salinivibrio sp. IB282

UPF0149 NTase_sub_bind Y1_Tnp RelE HTH_3 YrbL-PhoP_reg Polbeta

NZ_FOTO01000022.1:4634-22740
Desulfomicrobium norvegicum

Exosortase_EpsH DUF3485 Glycos_transf_4 TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs PhdYeFM_antitox PIN RelE HTH_3 ? ? Patatin DUF1538 HTH_25 Prok-E2_A ThiF Prok-JAB PilZ Phage_integrase

NZ_BMXH01000019.1:8084-20003
Aidingimonas halophila

HTH_1 LysR_substrate Amidohydro_1 HUTI_composite_bact TauE HTH_3 RelE 2-Hacid_dh_C DHDPS MR_MLE_N MR_MLE_C TctA TctB TctC Xan_ur_permease FMN_dh

NZ_MWZE01000122.1:3311-5931
Rickettsia endosymbiont of Culicoides newsteadi

UPF0102 Fe-S_biosyn NifU_N Transposase_31 HTH_3 RelE MFS_1 Aldedh

NZ_LN890655.1:2307011-2322990
Candidatus Promineifilum breve

Arginase DS SBP_bac_8 TOBE_2 ABC_tran AFOR_N AFOR_C RelE NTP_transf_2 HTH_19 DUF2442 DUF2442 ? Hydrolase CAP BPD_transp_1 BPD_transp_1

NZ_OCNJ01000008.1:135626-150650
Caenispirillum bisanense

LysM ? DNA_pol_A DNA_pol_A_exo1 5_3_exonuc 5_3_exonuc_N zf-CHCC ABC_tran DrrA1-3_C Hydrolase_4 HTH_3 NAD_Gly3P_dh_N NAD_Gly3P_dh_C EndIII_4Fe-2S HhH-GPD Guanylate_cyc DUF4139 DUF4140 RmuC RelE Epimerase TerC

NZ_POQS01000002.1:24814-38675
Achromobacter pulmonis

? RadC DUF5983 DUF2442 AAA_23 DUF6088 HTH_3 RelE MazE-like PemK_toxin ? DUF1173 T4SS_pilin Gp3-like Phage_integrase

NZ_NHRI01000002.1:320184-329082
Shewanella sp. BC20

? ? ? ? Ogr_Delta HTH_3 RelE Bro-N ? RNase_T ? ? ? ? Toprim_2

NZ_QGLF01000002.1:193462-207926
Zavarzinia compransoris

ADH_zinc_N ADH_N Abhydrolase_6 Phage_integrase HTH_3 HTH_3 OMP_b-brl ? AAA_25 Prim-Pol HTH_17 ? ? AAA_25 ? RelE Gp49

NZ_UGSM01000004.1:497-7463
Pantoea agglomerans

? ? ? HTH_3 ? ? ? ? ? ? RelE

NZ_JAMFSB010000106.1:15570-27728
Paraburkholderia sp.

PhyR_sigma2 Sigma70_r4_2 zf-HC2 GTP_cyclohydro2 CBS CBS DUF4148 BON HTH_3 RelE Ni_hydr_CYTB Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N Pur_DNA_glyco NAD_binding_10 HTH_18 ? Catalase

NZ_CAAAHX010000014.1:38736-52891
Legionella gresilensis

AOX Acetylase His-Kinase His-Kinase Ank ANKs Trans_reg_C Receiver ? ? RelE MqsA_antitoxin Glyoxalase Resolvase ABC_membrane ABC_tran DUF4156 PAS ?

NZ_JACDTY010000032.1:50453-10669
Mesorhizobium neociceri

SCO4226-like MqsA_antitoxin RelE ? ? SecA_DEAD P-loop_SecA SecA_SW SEC-C Rotamase_3 TetR_N TetR_C_11 GerE AAA_16

NZ_CP041730.1:614780-627565
Chitinimonas arctica

HTH_20 DUF948 ? ? LRRs Peptidase_M23 SH3_3 Receiver DeoRC HTH_DeoR DAO DAO_C ABC_tran OB_MalK TOBE_2 ABC_tran OB_MalK TOBE_2 ? CsbD RelE HTH_3

NZ_BSPF01000010.1:20289-34531
Mesorhizobium tianshanense

TrbC T2SSE Autoind_synth MerR ParA-Soj-NTPase ParB_N RepB RP-C RP-C_C Resolvase HTH_3 ? ? Phage_integrase Phage_int_SAM_1 SMC_ScpB AbiEii RelE ?

NZ_BKAG01000056.1:3065-16693
Brevifollis gellanilyticus

UPF0261 UPF0261_C ADH_N ADH_zinc_N Sigma70_r2 Sigma70_r4_2 ? Plug HTH_3 PNPOx_N Aminotran_1_2 SASA Queuosine_synth ? Metal_resist DUF218 Sulfatase_C Sulfatase RelE

NZ_LMUJ01000079.1:133705-146713
Acidisphaera sp. S103

Amino_oxidase ABC_tran ? ? ? ? RelE HTH_31 Receiver ? ? LysR_substrate HTH_1 HUTI_composite_bact ? PP-binding AMP-dom_DIP2-like

NZ_JAASQO010000010.1:171423-187473
Parvibaculum indicum

? ? Methyltransf_25 B12-binding MM_CoA_mutase ? Abhydrolase_6 GerE AurF HTH_3 ADH_N ADH_zinc_N TrkH Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 HIG_1_N ? PagL RelE

NZ_BAABGB010000054.1:10514-25415
Gluconacetobacter asukensis

P-II Ribosomal_S9 Ribosomal_L13 RHH_1 MFS_3 ? ? HTH_31 Aminotran_1_2 Gln-synt_C Gln-synt_N HhH-GPD ADH_zinc_N ADH_N Abhydrolase_1 RelE ?

NZ_JACHIK010000020.1:61748-75132
Shinella fusca

APH RNase_H Redoxin DsbD_N DUF179 CsbD RelE MqsA_antitoxin Acetylase DUF1127 Peptidase_M16_C Peptidase_M16 THR4_C PALP Thr_synth_N Zn_ribbon_DZR_6 Peptidase_Mx ? ?

NZ_BMJA01000002.1:1019906-1034821
Dyella nitratireducens

Abhydrolase_6 Cupin_2 Abhydrolase_1 OsmC AhpC-TSA Receiver Trans_reg_C RelE HTH_3 ? Receiver His-Kinase His-Kinase Y_Y_Y Reg_prop Reg_prop Reg_prop Aminotran_1_2 GntR EamA EamA ? HAMP His-Kinase His-Kinase PadR

NZ_JACXAB010000069.1:11209-8410
Microvirga sp.

? ? HTH_3 ? ? DDE_Tnp_IS66 zf-IS66 DUF6444 RelE Guanylate_cyc Usp HTH_Crp_2 cNMPBD DDE_Tnp_IS240

NZ_JACYLH010000002.1:143002-150405
Cysteiniphilum marinum

? ? DUF29 DUF29 DUF29 DUF2442 DUF4160 RelE MqsA_antitoxin KilA-N CsrA ? DUF1351 YqaJ RecT

NZ_JAFIDN010000005.1:241311-256492
Natronogracilivirga saccharolytica

? ? TonB_dep_Rec_b-barrel Plug 2-Hacid_dh_C ACT tRNA-synt_1c_C2 tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c Phage_integrase Phage_int_SAM_4 RelE HTH_3 HTH_3 PhdYeFM_antitox ParE_toxin AAA_33 ? DDE_Tnp_ISL3 EB_dh

NZ_CP076853.1:2218760-2232718
Rhizobium sp. WYJ-E13

DUF1223 Ni_hydr_CYTB Oxidored_molyb adh_short_C2 TetR_C_13 TetR_N MqsA_antitoxin RelE MFS_1 TetR_N ADH_zinc_N_2 ADH_N MFS_1 Glyoxalase adh_short_C2 SGSH_C Sulfatase

NZ_JAHUWX010000003.1:193526-200391
Acidovorax sp. sif0715

? ? ? ? ? ? HTH_3 RelE ? HicB ? ? ? ? HicB

NZ_CP073764.1:2202657-2218075
Methylovirgula sp. HY1

RBFA ? GTP_EFTU_D2 IF-2 EF-G_D2 GTP_EFTU IF2_N IF2_assoc KH_NusA_C KH_5 S1 NusA_N RelE MqsA_antitoxin Arginosuc_synth Arginosuc_syn_C SWIB PGI ? YkuD AMP-binding_C AMP-binding ACAS_N ? YlxR DUF150_C RimP_N

NZ_JAKLWS010000009.1:140317-150142
Rhodohalobacter sulfatireducens

? DUF4386 ADH_zinc_N_2 ADH_N FA_hydroxylase Dabb UPF0175 DUF3368 RelE HTH_3 ? ? ? ? PhdYeFM_antitox

NZ_JAKSXU010000051.1:425-4936
Methylobacterium sp. J-092

? ? ? Bac_DNA_binding HTH_3 GntR ? RelE ?

NZ_JALKCH010000007.1:279537-287215
Ancylobacter crimeensis

KdpC His-Kinase His-Kinase GAF_3 DUF4118 KdpD Trans_reg_C Receiver ? ? ? MqsA_antitoxin ? RelE

NZ_JAHHPO010001172.1:154-2277
Endozoicomonas sp. ONNA2

HTH_3 RelE HTH_3 HigB-like_toxin ? Ogr_Delta

NZ_CP096972.1:27381-43734
Sphingomonas sp. SUN039

SnoaL_4 Pyr_redox_2 Rieske Ring_hydroxyl_A TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C Aldedh ADH_N ADH_zinc_N SnoaL_2 RelE HTH_31 SGL SGL IclR_C HTH_IclR AMP-binding AMP-binding_C FAS1_DH_region FA_hydroxylase SnoaL_4

NZ_CP110921.1:1906460-1916056
Methylococcus mesophilus

? ? ? HTH_17 HTH_19 SLATT ? RelE HTH_31 YdjM ? Excalibur SNase SRAP ? ?

NZ_JAPMLN010000029.1:6647-8399
Marinicella gelatinilytica

AAA_15 AAA_15 OLD-like_TOPRIM Methylase_S Methylase_S HsdM_N N6_Mtase HTH_3 RelE Amidohydro_1

NZ_JAPZEI010000004.1:341264-354934
Coraliomargarita parva

Peptidase_M90 DEAD Helicase_C VF530 zf-CDGSH TetR_C_13 TetR_N OsmC HTH_3 RelE GatB_N GatB_Yqey PDDEXK_3 Amidase GatC tRNA_bind B3_4 B5 tRNA_synthFbeta FDX-ACB Phe_tRNA-synt_N tRNA-synt_2d ?

NZ_DALYQB010000031.1:25568-6736
Sphingobium yanoikuyae

Phage_portal TerL_ATPase TerL_nuclease ? ? ? ? RelE HNH Mom

NZ_CAJYMR010000007.1:280872-294406
uncultured Pseudomonas sp.

Phage_tail_terminator_5 Phage_tail_3 Phage_tail_3 Phage_TAC_4 ? HTH_3 RelE ? DUF4214 DUF4214 Gly-zipper_Omp Tape_meas_lam_C TMP_2 ? ? DUF4376 ? Phage_lysozyme DUF1799

NZ_JADGEW010000001.1:460164-471576
Amphritea sp.

? TetR_N TetR_C_13 TM_HPP MarR 4HBT_3C 4HBT_3 YggL_50S_bp HTH_3 RelE ? PDDEXK_5 ? SmpB SNF SNF Polyketide_cyc Ub-RnfH

NZ_DDFB01000010.1:54023-10581
Novosphingobium sp. UBA1939

WYL TerB Caspase TIR ? HNH RelE HTH_3 ? Phage_GPD Phage_P2_GpU DUF3511 PhageMin_Tail Phage_P2_GpE Phage_TAC_7

NZ_CAJUIG010000196.1:1-5407
uncultured Bilophila sp.

REase Methylase_S Methylase_S HsdM_N N6_Mtase ? HTH_31 RelE ?

NZ_CALMSV010000220.1:1-11441
uncultured Sphingorhabdus sp.

DUF3558 HTH_1 LysR_substrate DUF4189 RelE ? ParA-Soj-NTPase P63C LepA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD DUF1289 AAA_28

NZ_JAUXNC010000022.1:1279-5000
Methylotenera sp.

? Trypsin RelE MqsA_antitoxin ? ? Polbeta HEPN DUF1016_N

NZ_JAUVWH010000014.1:20162-33902
Geoalkalibacter sp.

His-Kinase His-Kinase HAMP sCache_4 ABC_sub_bind ? Oxidored_nitro Oxidored_nitro NifU_N NifQ RelE HTH_3 Pribosyltran Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 Peptidase_M23 LysM VTT_dom PCMT SurE Sod_Fe_C Sod_Fe_N

NZ_JAUSVS010000007.1:104766-121026
Caulobacter ginsengisoli

? ? ? ? ? DUF6035 DUF7830 ? RelE HTH_3 ? Hjc HHH_5 HTH_61 Helicase_C DEAD ? ArdcN MPTase-PolyVal Relaxase MobC

NZ_DAMABV010000002.1:346910-363149
Phenylobacterium sp.

? ? ? HTH_63 TIR ? HTH_3 RelE HTH_3 DUF3644 ArdA TraG-D_C ? ? Relaxase

NZ_JAUGZK010000002.1:126022-138538
Alkalimonas mucilaginosa

gag-asp_proteas Amidase_2 AmpE Transketolase_N Transketolase_N PDH_E1_M Transketolase_C_1 Biotin_lipoyl Biotin_lipoyl E3_binding 2-oxoacid_dh Pyr_redox_2 Pyr_redox_dim RelE HTH_3 Acetylase Polbeta ROF 4HBT_2 NTase_sub_bind GntR FCD MgtC

CP141262.1:1643509-1659821
Luteimonas sp. R10

Methyltransf_23 Methyltransf_23 Glyco_transf_4 Glycos_transf_1 ? N_methyl Pilin N_methyl Pilin MqsA_antitoxin RelE Receiver NtrC-AAA AAA_lid_14 HTH_8 6TM_PilS PAS His-Kinase His-Kinase Ligase_CoA ATP-grasp_2 Ligase_CoA CoA_binding DUF559 NAD_synthase CN_hydrolase VKOR